MERCH
PRODUCT ANALYTICS

EXPLORE THE STORY

- 01 Overview
- 02 Customers
- 03 Discovery
- 04 Evidence

THE NEXT MOVE

Validate tracking.
Test discovery.

GOOGLE MERCHANDISE STORE

Historical GA4 sample
Nov 2020 - Jan 2021

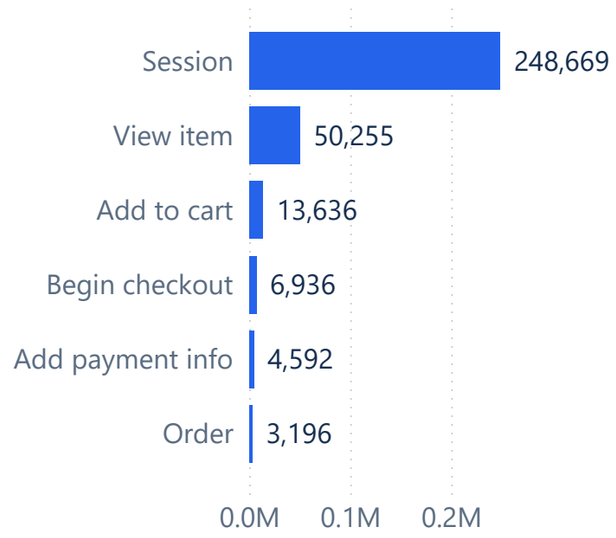
The purchase journey

Where buyers are lost, what the opportunity is, and where to focus | 26 Nov 2020 - 25 Jan 2021

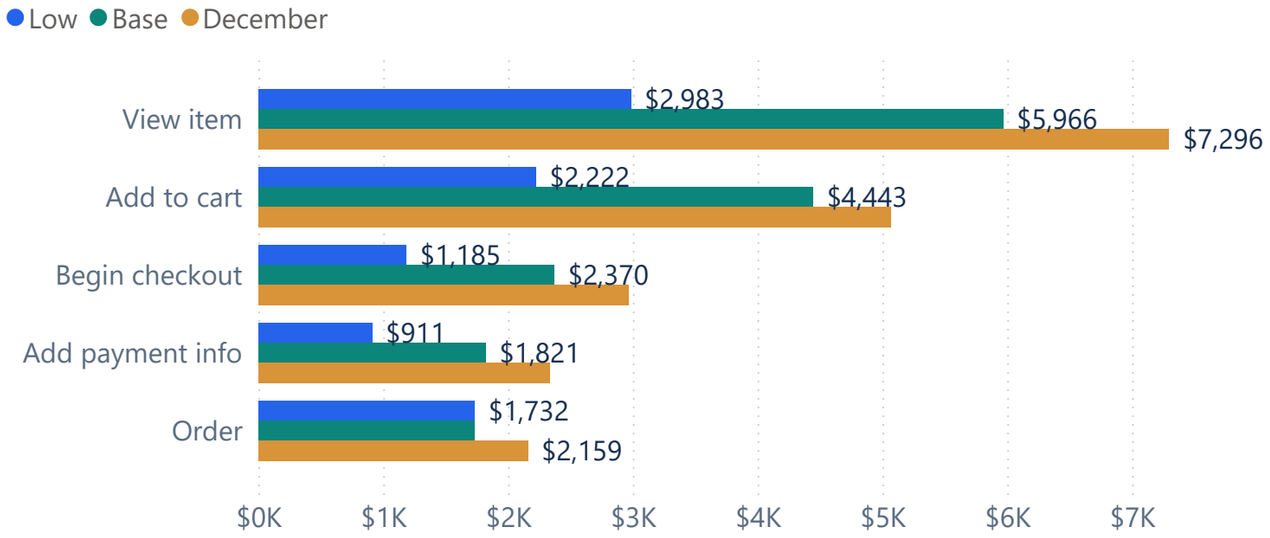
Sessions analysed	Valid orders	Session conversion	Average order value
248,669	3,540	1.29%	$69.25

THE OPPORTUNITY / Four in five sessions record no product view. Validate tracking, then test discovery for new visitors.

01 Sessions by purchase step



02 What could +1 percentage point be worth?



USD per 30 days. Low/base/December are business scenarios, not statistical intervals.

03 Which audience needs attention?

Compare by: New vs returning

Audience	View item	Add to cart	Begin checkout	Add payment info	Order
New	18.5%	23.3%	44.3%	51.1%	66.7%
Returning	25.3%	35.4%	60.0%	81.9%	71.5%

INTERPRETATION / Funnel counts event presence within a session, not strict sequence. Orders and purchasing sessions are different units.

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Customers, intent & the next test

Understand who returns, what predicts purchase, and how to size the discovery experiment

2.02x adjusted odds

First-session add to cart predicts later purchase

29,259 per arm / 3 weeks

To detect a 5% relative product-view lift

01 Signals of a later purchase

First-session signal	Odds ratio	CI low	CI high
Added to cart in first session	2.02	1.56	2.60
Began checkout in first session	0.56	0.41	0.78
Device: mobile	0.93	0.77	1.13
Device: tablet	0.75	0.37	1.54
First-touch channel: Organic Search	1.07	0.84	1.37
First-touch channel: Paid Search	1.22	0.79	1.87
First-touch channel: Referral	0.96	0.70	1.33
First-touch channel: Unknown	1.05	0.78	1.41
log(1 + engagement seconds)	1.88	1.75	2.03
Market: India	1.11	0.78	1.56

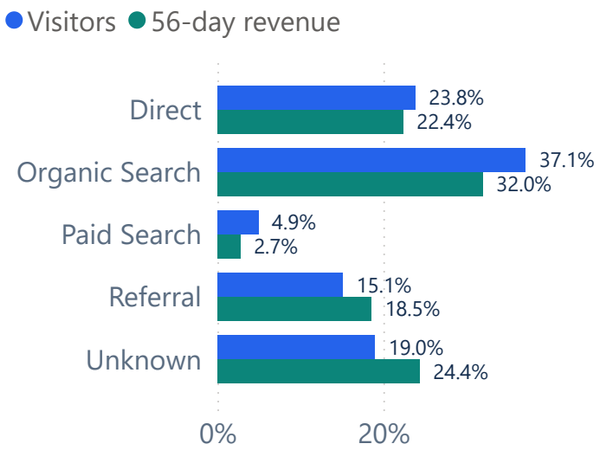
First-session non-buyers, followed for 30 days. Adjusted odds ratios with 95% intervals; predictive, not causal.

02 Do new visitors return?

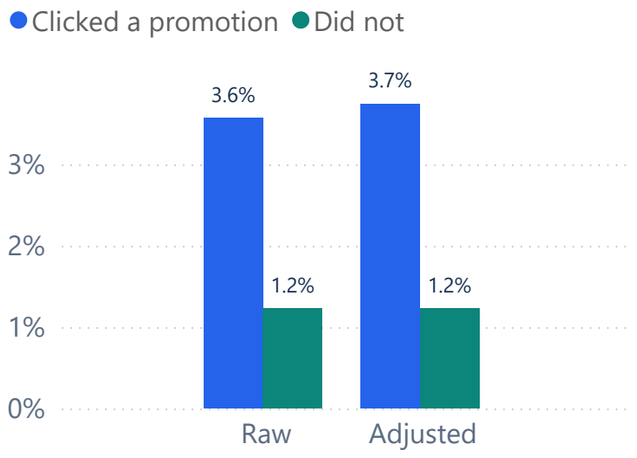
First week	1	2	3	4	5	6	7	8
26 Oct 2020	6.1%	1.6%	0.9%	0.5%	0.4%	0.2%	0.1%	0.2%
02 Nov 2020	4.2%	1.9%	1.4%	1.2%	1.0%	0.7%	0.2%	0.2%
09 Nov 2020	5.7%	2.4%	1.7%	1.8%	0.9%	0.5%	0.2%	0.4%
16 Nov 2020	5.2%	2.8%	2.0%	1.5%	0.5%	0.3%	0.3%	0.3%
23 Nov 2020	4.7%	2.2%	1.3%	0.6%	0.4%	0.4%	0.4%	0.3%
30 Nov 2020	5.0%	2.1%	0.8%	0.6%	0.6%	0.4%	0.3%	0.3%
07 Dec 2020	3.1%	0.9%	0.6%	1.0%	0.5%	0.3%	0.3%	
14 Dec 2020	2.8%	0.9%	1.2%	0.9%	0.5%	0.4%		
21 Dec 2020	2.5%	1.1%	1.0%	0.6%	0.4%			
28 Dec 2020	2.4%	0.9%	0.6%	0.4%				

Weekly cohort activity. Empty cells can reflect no recorded activity or incomplete follow-up.

03 Acquisition quality



04 Promotions & conversion



05 Experiment sizing

Target lift	Users per arm	Weeks needed
2.0%	180,790	16
3.0%	80,660	7
5.0%	29,259	3
7.5%	13,126	2
10.0%	7,451	2

INTERPRETATION / Channels are first-touch; promotions are observational. Test sizing: 80% power, two-sided alpha 5%, equal allocation.

Inside the discovery gap

Observed landing pages, product-view coverage and where non-viewers finish | Funnel window

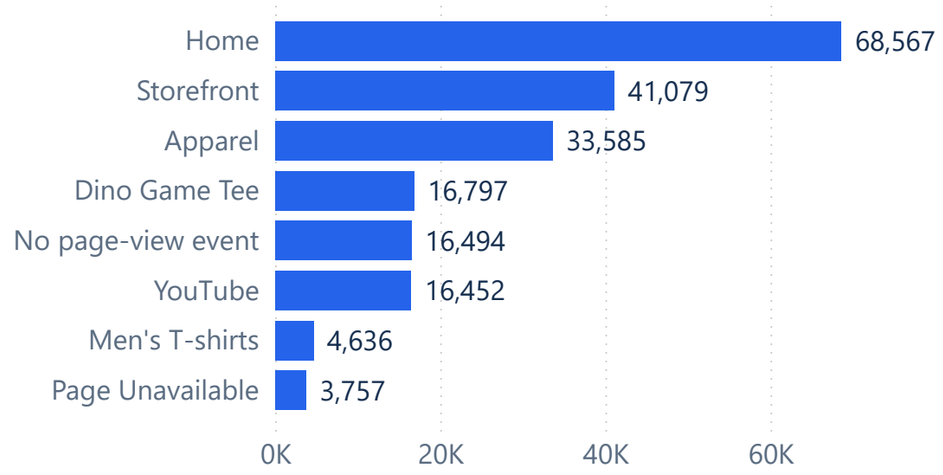
79.8%
Sessions without a product-view event

6.6%
Sessions without any page-view event

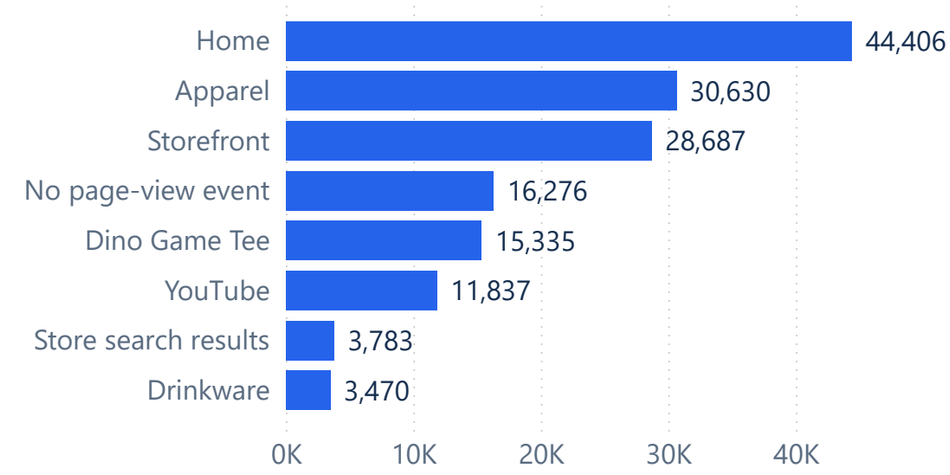
27.6%
Sessions first seen on Home

BEFORE TESTING / Product-titled pages sometimes lack view_item. Confirm instrumentation before treating missing views as navigation failure.

01 Where do sessions begin? | Top 8 titles



02 Where do non-viewers finish? | Top 8



03 Compare product-view coverage across landing pages

Observed landing title	Landing sessions	Records a product view	Session order rate
Home	68,567	24.2%	2.43%
Storefront	41,079	12.3%	0.57%
Apparel	33,585	7.5%	0.24%
Dino Game Tee	16,797	7.5%	0.01%
No page-view event	16,494	1.3%	0.00%
YouTube	16,452	19.5%	0.21%
Men's T-shirts	4,636	35.7%	0.67%
Page Unavailable	3,757	5.4%	0.00%

INTERPRETATION / First/last recorded page_view titles are not unique URLs or verified acquisition pages. Absence of an event is not proof of intent.

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Trust the evidence, bound the claim

Tracking coverage, sampling uncertainty and sensitivity to analytical definitions

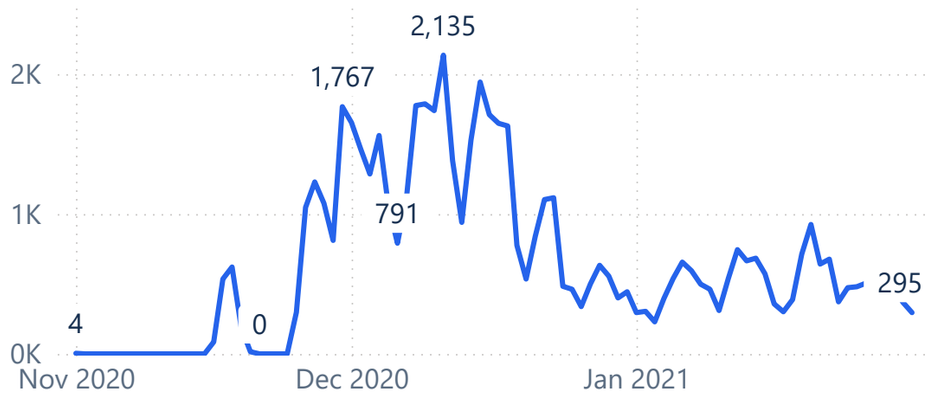
61 reporting days
26 Nov 2020 - 25 Jan 2021 funnel window

92 source days
Daily audit preserves the full raw period

1,000 resamples
Whole users, with all their sessions retained

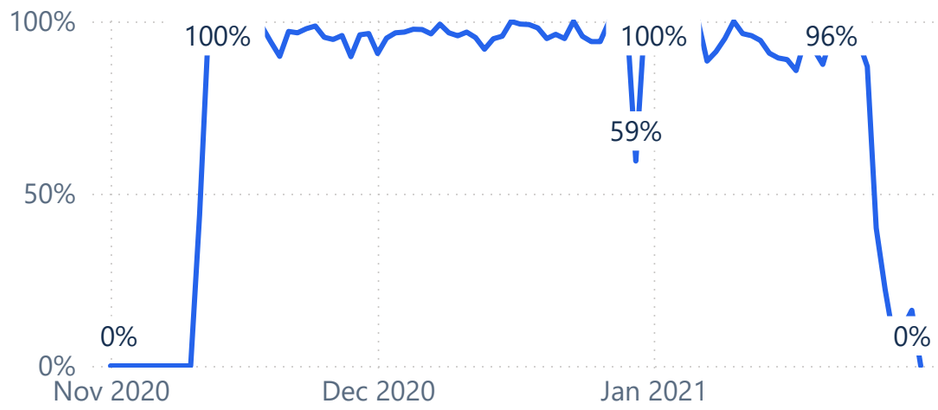
CONFIDENCE WITH LIMITS / Discovery leads under the +1-point model. Its sampling interval does not establish causal revenue impact.

01 Daily add-to-cart tracking



Latest: 31 Jan 2021 | 295 cart events

02 Purchase events with valid order IDs



Latest: 31 Jan 2021 | 0% valid order IDs

03 Sampling uncertainty in the scenario

Stage	USD / 30d	95% low	95% high
View item	$5,966	$5,641	$6,295
Add to cart	$4,443	$4,192	$4,693
Begin checkout	$2,370	$2,243	$2,498
Add payment info	$1,821	$1,723	$1,917
Order	$1,732	$1,638	$1,823

04 Do definitions change the picture?

Definition	Orders	Conversion	View rate
Baseline	3,540	1.29%	20.2%
Order key: user + transaction	3,555	1.29%	20.2%
Require session_start	3,504	1.29%	20.1%
Start 25 November	3,635	1.29%	20.3%

INTERPRETATION / The audit misses partial recovery on 16-19 and 25 Nov. The 26 Nov start also relies on daily inspection; bootstrap CIs are conditional.